WHITE PAPER EXCERPT

Predicting High-Risk Non-Sufficient Funds

Excerpt — Sections 1–2: Objective & Background

STOCHOS.

About this excerpt. This is Sections 1 (Objective) and 2 (Background) from a longer internal research paper originally developed for a state lottery client, reproduced here to illustrate the business problem and operational context behind our analytical approach. Some figures, counts, and internal system names referenced elsewhere in the paper have been generalized or altered to protect client confidentiality — the context and reasoning are represented faithfully.

1. Objective

To research a methodology to predict and prevent the occurrence of high-risk non-sufficient funds instances to reduce risk and prevent loss for a Lottery and its retailers.

2. Background

A Lottery generates revenue by selling lottery games through approximately 18.8 thousand retailers. Each week, Lottery retailers are invoiced, and the funds for payment are swept and remitted to the Lottery. Sweeps occur on Wednesday for Draw products and Thursday for Scratchers products. When a retailer does not have adequate funds in their bank account for the weekly sweep, a Non-Sufficient Funds (NSF) instance occurs. The Lottery's internal bank processes weekly retailer account sweeps and notifies the Lottery via a file uploaded to Lottery databases that contain retailers that have incurred an NSF. In addition, the Retailer Financial Services (RFS) unit dispatches collectors that call retailers to inquire about the NSF. While a retailer has not paid their obligation on inventory, the Lottery will typically inactivate the retailer's Lottery terminal, temporarily halting any sales activity. The retailer may pay the NSF by check mailed to the Headquarters building or by paying the nearest District Office. In addition, a Dunning Letter is mailed to the retailer every ten days, and sales activity remains halted until the

Lottery receives compensation. The Lottery terminates business with the retailer after the third Dunning Letter. Team members will generally retrieve all unsold Lottery products and Lottery equipment.

NSFs represent an infrequent occurrence at the Lottery, but a portion of the NSF's ultimately end up as lost revenue, accounting for losses of approximately $36 million from FY 2015–20. During that time, the Lottery conducted 6.2 million sweeps with NSFs occurring on 8,568 of those instances (.14% of the time). Out of the nearly $20.6 billion swept, $31.45 million returned as NSF (.15% of total dollars swept). The Lottery collected roughly 83.5% of the NSFs incurred in FY 2019–20.

Though NSFs occur infrequently and are largely collected on, understanding and predicting the outcome of an NSF does pose an opportunity to mitigate risk and prevent loss. Each NSF is a resource encumbrance when an NSF occurs. Staff key the NSF into the Lottery's financial system, and a collector escalates the collections process by directly communicating with the retailer. A District Sales Representative (DSR) may need to visit the location. Further, when a retailer is inactive for any duration, they are not selling Lottery products. Retailers that NSF that fail to remit payment result in a loss to the Lottery, as the unpaid NSF eventually is processed as bad debt. Lastly, an NSF that leads to termination may represent a loss to the Lottery in future earnings.

2.1 Philosophy of an NSF

An NSF is an adverse occurrence for the Lottery and a retailer. However, NSFs, by nature, are not inherently wrong as they are a function of doing business at the Lottery. The way to reduce the NSF rate to zero is to terminate all retailers or generate no revenue, so the Lottery understands that an NSF rate is an accepted part of the Lottery business. The optimal goal is to a) understand an acceptable NSF rate for the Lottery to operate with (whether globally, by trade channel, by District Office, or some combination thereof), b) understand NSFs that are preventable (and NSFs that are not preventable), and c) understand and limit the occurrence of NSFs that may be detrimental for the Lottery.

NSFs occur for the following reasons, and an assessment of the whether the NSF is preventable or not is listed below:

Table 1: NSF Reasons and Preventability Table

NSF Type	Preventable
Retailers taking advantage of our NSF policy	No
Retailers committing fraud	Yes
Retailers improperly managing finances	No
Retailers incurring inventory saturation	Yes
Retailers experiencing long-term issues	Yes
Bank errors	No

The Lottery cannot prevent retailers who NSF in all instances, as many instances are singular events. Further, the Lottery cannot prevent retailers who knowingly abuse the Lottery's NSF terms. Currently, retailers incur an NSF fee of $25 for each NSF, which has been treated as a "$25 short-term loan". Lastly, the Lottery cannot predict bank errors or frozen accounts. There will always be a challenge trying to build a predictive model to predict the occurrence of an NSF for all retailers as no internal data may lend to the predictability of singular instances from occurring.

The Lottery can work towards predicting some NSFs from occurring, however. Retailers who are potentially scratching their own Scratchers product and retailers experiencing inventory saturation may provide a signal. Lastly, retailers who experience long-term issues may be at higher risk for multiple NSFs or termination.

2.2 High-Risk NSFs

There are NSFs that are detrimental to the Lottery, even if an NSF, by nature, is not inherently bad. In most instances, the Lottery receives prompt payment for an NSF. However, when the NSF payment is not prompt, collections efforts require more Lottery resources. Further, these types of NSFs result in terminal deactivation and possibly termination. The NSF may eventually be accounted for as bad debt, an expense to the Lottery, and is ultimately treated as a loss.

The following table details the median count, median NSF by amount, and median count of days since the last NSF for NSFs by days to pay.

Table 2: NSF Details by Days to Pay

Name	Count	Med NSF	Med Days Since Last
NSF less than 10 days	3,953	$1,574	147
HRNFS over 10 days	767	$1,513	63
HRNSF over 20 days	281	$1,399	21
HRNSF over 30 days	1,368	$2,331	21
Total	**6,370**	**$1,689**	**42**

NSF instances that are paid within 30 days or less share similar median amounts, however the days since the last NSF indicate that the longer it takes to pay an NSF, the higher probability the retailer may NSF again.

As NSFs are paid later, the risk of termination increases. The research team viewed unique retailers that NSFed compared to terminations that occurred within 60 days. The rate steadily increases the later NSFs are paid. When a retailer takes longer than 30 days, they experience a high likelihood of termination.

Table 3: Termination Rates by NSF Days to Pay

Name	Retailer Count	Term Less Than 60 days	Percent
NSF less than 10 days	2,027	162	7.99%
HRNFS over 10 days	586	81	13.82%
HRNSF over 20 days	217	50	23.04%
HRNSF over 30 days	569	533	93.67%
Total	**2,582**	**825**	**31.95%**

NSFs that are paid in less than 30 days indicate a retailer that still wants to be part of the Lottery family. Retailers who take longer to pay NSFs may be placed on probation, but they are still likely to remain a Lottery retailer. Retailers who take longer than 30 days to pay their NSF end up paying the NSF in full.

Table 4: Collection Rates by NSF Days to Pay

Name	Total	Collected	Outstanding	Percent
NSF less than 10 days	$14,750,934	$14,750,934	$0	100%
HRNFS over 10 days	$2,669,286	$2,669,286	$0	100%
HRNSF over 20 days	$2,507,183	$2,507,183	$0	100%
HRNSF over 30 days	$6,277,467	$3,726,679	$2,550,788	59%
Total	**$26,208,868**	**$23,658,082**	**$2,550,786**	**90%**

The research team categorized all NSFs over 10 days to pay as an HRNSF. HRNSFs represent a major risk to the Lottery because they represent retailers at a higher risk of incurring another NSF relatively quickly, are likelier to lead to termination, and offer less chance of repayment.

2.3 Resource Allocation for NSFs

When a retailer incurs an NSF the Retailer Financial Services unit is responsible for getting in contact with them and getting a payment made as fast as possible. The process is performed by a series of Program Tech IIIs, also known as collectors, and can be divided into 5 steps: populating the NSF list, contacting the retailer, keying in the NSF to Epicor, emailing the NSF list to Sales, and assisting the retailer with payment.

To start the process, a collector populates an NSF report each day that details any retailer who incurred an NSF the day prior. The report is then divided amongst 3 collectors equally who begin contacting the retailer for payment. Additionally, a collector emails this daily report to Sales so that the District Sales Representative (DSR) are aware of any retailer on their route that incurred an NSF. Once a collector has contacted the retailer, they work on scheduling the sweep of their account for the missed payment. The table below details the resources used for each type of NSF.

Table 5: Resources Used by NSF Type

NSF Type	Populate List	Contact Retailer	Key in NSF	Email to Sales	Assist Retailer	Hours Used
NSF less than 10 days	2	2.5	0.5	1	2	8
HRNFS over 10 days	0	6	0	1	2	14
HRNSF over 20 days	0	6	0	1	2	20
HRNSF over 30 days	0	6	0	1	2	26

As payments are prolonged, more resource time is used for each type of NSF. If the retailer does not provide payment within 10 days, a Dunning letter is issued. The collectors continue to attempt contact with the retailer for payment, and if successful, will assist the retailer in providing payment. Each 10 days a new Dunning letter is mailed to the retailer, and after 3 Dunning letters a 4th Dunning letter is issued and the retailer is terminated.

The 4 collectors, on average, are paid $8,618 a month, including salary and benefits. At an average hourly rate of 176 hours a month, this equates to $48.97 an hour.

Table 6: Resource Cost by NSF Type

NSF Type	Hours	Rate	Total Resource
NSF less than 10 days	8	$48.97	$391.76
HRNFS over 10 days	14	$48.97	$685.58
HRNSF over 20 days	20	$48.97	$979.40
HRNSF over 30 days	26	$48.97	$1,273.22

The longer it takes a retailer to pay their NSF, the more resource time and cost is allocated to the NSF instance. Further, resource time for each NSF may vary. Retailers may have multiple contacts with multiple phone numbers, and sometimes the contact information is incorrect or out of date. Further, retailers may need assistance with payment or understanding their invoice. The longer each NSF takes to be collected on, the more resource time is required to perform the contact attempts for each retailer. HRNSFs stress the time and cost of the RFS unit to continually attempt contact with retailers for payment.

2.4 Inventory Saturation

An NSF occurs when a retailer lacks adequate funds in their bank account to satisfy the most recent invoice during a Lottery sweep; this happens for various reasons, such as a retailer mismanaging their finances, retail employee embezzlement, or a retailer abusing the Lottery's NSF terms. This responsibility falls mainly on the retailer to provide the correct payment. However, one instance where the Lottery may play a part in exacerbating the severity of NSFs may be, in part, from inventory saturation. The Lottery controls inventory distribution at retail. If the Lottery is inundating a retailer with inventory, this may compound the occurrence and gravity of HRNSFs. A Scratchers pack can settle (and become due for payment) in various ways, namely due to selling (when a Scratchers pack reaches a certain validation point where the pack automatically settles and is due for payment) and also due to time. Pack information, invoice information outside of historical bounds, and sales averages may offer insight into whether the Lottery is potentially causing an NSF due to inventory saturation.

2.5 COVID Pandemic

Due to the COVID-19 pandemic, settlement terms were temporarily adjusted to allow retailers more time to sell packs during the pandemic. Since January 2021, retailers have 150 days until a pack settles and must be paid for. With this change to settlement terms, packs are no longer settling as quickly, which is causing retailers to NSF less. This is artificially reducing the total amount of NSFs, however once the Lottery returns to normal settlement terms the NSF rate is expected to normalize to prior levels.

In January 2021, the Lottery decided to temporarily cease the $25 fee applied to retailers that NSF. The change was enacted to provide further relief for retailers during the pandemic. Prior to the pandemic, the Lottery would collect, on average, $57,000 in fees annually. The Lottery is expected to reinstitute the fee in 2023.

This excerpt covers only the Objective and Background sections of the full paper. The complete research paper also includes detailed methodology, findings on model performance and variable importance, fiscal impact analysis, and operational recommendations for deployment.